

Analog CMOS IC Design (EE651)

Assignment- 2

Total Marks – 15 points

Date: 22/08/2025

Submission Date: 29/08/2025 (11:55 PM)

Note: Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

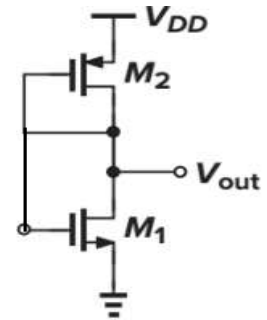
- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
- Name the report file as: RollNumber_Assignment2.pdf

1. **Voltage divider circuit** built using MOS transistors shown in figure. V_{out} is a function of W/L , C_{ox} and $V_{DD} = 1.8$ V. [5]

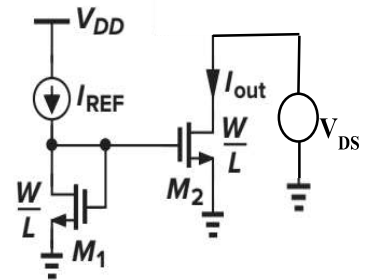
- a. Design the circuit to achieve V_{out} values of $0.25V_{DD}$, $0.5V_{DD}$, and $0.75V_{DD}$.
- b. For the $V_{out} = 0.5V_{DD}$ design, plot V_{out} as a function of temperature (-40 °C to 150 °C) for different process corners (FS, FF, TT, SF, and SS).



Explain the behaviour observed in the plots.

2. Design the current mirror circuit as given below, with I_{REF} (ideal element) of 15 μ A and desired I_{out} of 75 μ A. Remember to choose W/L of M_2 such that its V_{ov} is greater than 150 mV and choose L of transistors to minimize the channel length modulation effect. [3]

- a. Vary the voltage at the drain of M_2 from 0 to V_{DS} (DC Sweep Simulation) and plot the current ratio: I_{out}/I_{ref} .
- b. Determine the percentage error in I_{out} when V_{DS} is at the edge of saturation compared to $V_{DS} = 1.8$ V.
- c. Vary the temperature from -40 to 150 °C (temperature Sweep Simulation) and plot the current ratio: I_{out}/I_{REF} .

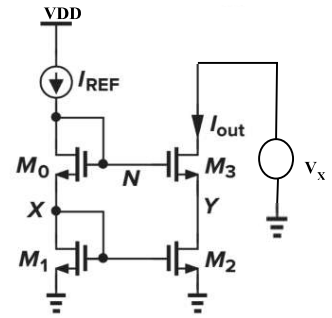


Explain your observations clearly.

3. Design the cascode current mirror circuit given below, with I_{REF} (ideal element) of 15 μ A and desired I_{out} of 75 μ A. [5pt]
 - a. Vary the voltage at the drain of M_3 from 0 to V_{DD} (DC Sweep Simulation) and plot the current ratio: I_{out}/I_{REF} . Explain the plot behaviour.

- b. Find the minimum voltage at the drain of M3 to ensure both M3 and M2 remain in saturation.
- c. Determine the percentage error in I_{out} when at the edge of saturation compared to $V_X=1.8V$.

Explain your observations clearly.



4. Compare Q2 (Simple Current Mirror) and Q3 (Cascode Current Mirror). [2pt]
 - a. Determine the output impedance for each.
 - b. Discuss advantages and disadvantages (pros and cons) of both circuits.

Report should include,

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary